

A conducting ring driven through an internal cut

Hybrid $A-\Phi$ finite-element solver, tree-cotree gauge. The first **multiply connected** case in the suite: the current circulates with no terminal anywhere on the outer boundary, so the ring can only be driven by cutting it.

regression_tests/05_Loop_1A_50Hz · 1 A internal current source ·
50 Hz · 524,050 unknowns · backward error 9.1e-23

0. What this case establishes

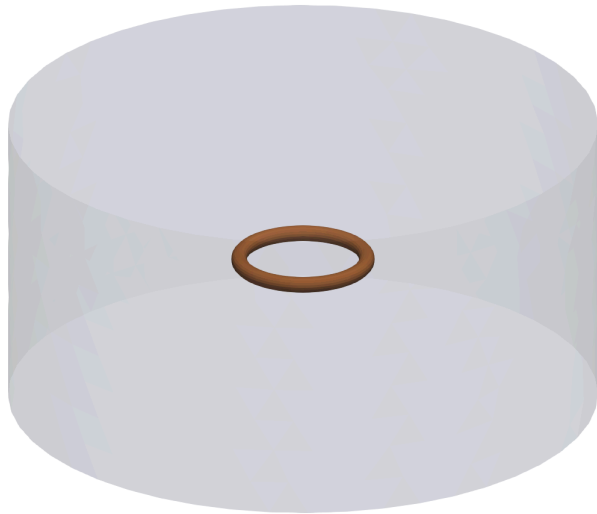
Cases 01–04 are all a straight rod. A rod is **simply connected** and the tree-cotree gauge has nothing hard to do on one. A ring is **multiply connected**: the cotree has to span the loop, the current has to be able to go all the way round, and there is no outer surface to attach an electrode to. It can only be driven through a **cut**.

The check no other case can make. Inject 1 A across the cut, then measure the current crossing a plane where *nothing is prescribed*:

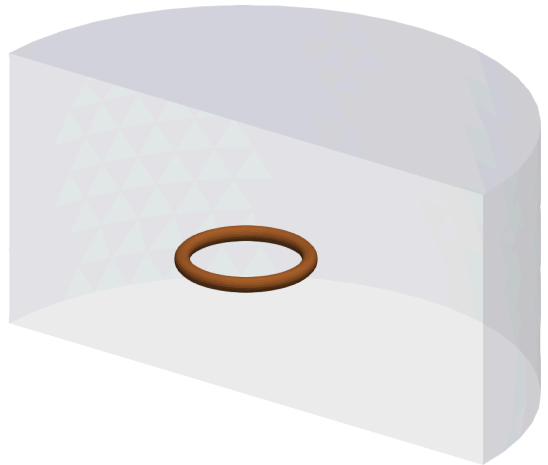
	MEASURED
I through the cut , where 1 A is injected	0.9999784 A
I through $\theta = \pi/2$, nothing prescribed	1.000183 A

The current circulates the whole way round. A gauge that wrongly killed the loop would show up there and in no other case in the suite.

1. The case



What was modelled. The copper ring at mid height inside the air cylinder, drawn transparent. Loop radius 6.5 mm and tube radius 0.8 mm against a domain 30 mm in radius and 30 mm tall — the ring is **small** in it, which is the point of §5.2's boundary test.



Cut at $y = 0$, the plane the port sits on, so the ring is visible rather than seen through the domain.

Conductor copper, $\sigma = 5.8e7 \text{ S/m}$; loop radius $R = 6.5 \text{ mm}$, tube circumradius $a = 0.8 \text{ mm}$, a 20-gauge wire

Domain cylinder of air, 30 mm radius, 30 mm tall, $\mathbf{n} \times \mathbf{A} = 0$ on the outer boundary

Excitation **1.0 A through the cut at $\theta = 0$** , 50 Hz

Skin depth $\delta = 9.346 \text{ mm}$, so $a/\delta = 0.086$ — no skin effect

Mesh 38,248 nodes, 237,031 elements, Netgen-optimised

Measured $Z = 3.543192e-04 + 6.087457e-06 j \ \Omega$

1.1 Where the dimensions came from

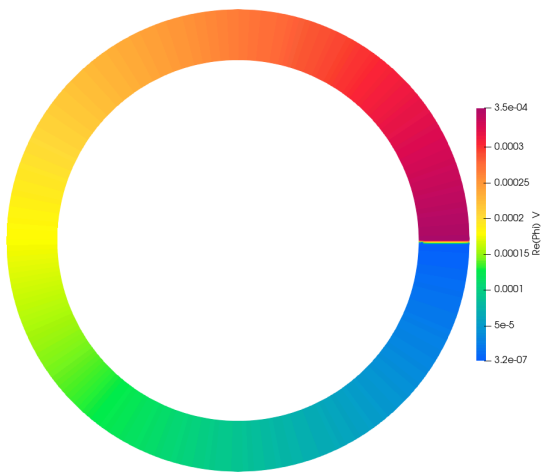
Not from a document. Two **published** closed forms, solved together against a reference impedance measured for this geometry:

$$\begin{aligned} R_{\text{dc}} &= 2 \pi R / (\sigma A) \\ L &= \mu_0 R \left[\ln(8R/a) - 2 + 1/4 \right] \quad \text{round wire, uniform current} \end{aligned}$$

$R = 6.5 \text{ mm}$ with $a = 0.8 \text{ mm}$ reproduces the reference R to 1.3 % and L to 0.27 %, and agrees independently with dimensions measured off the reference model's own scale bar — centreline diameter 1.23 cm, tube 1.4–2.0 mm.

The reference was run at 100 Hz, not 50. That was stated nowhere and falls out of the same two equations. $\text{Im}(Z)/\omega$ is 19.75 nH at 100 Hz and 39.50 nH at 50 Hz, and *no* geometry consistent with the reported R can give 39.50 nH. It does not matter for what this case checks: a/δ is 0.086 at 50 Hz and 0.121 at 100 Hz, so there is no skin effect either way and R and L are frequency independent between them.

2. The potential



Re(Φ) over the whole ring, seen from $+z$. It ramps smoothly all the way round and **falls off a cliff at the cut**, at 3 o'clock.

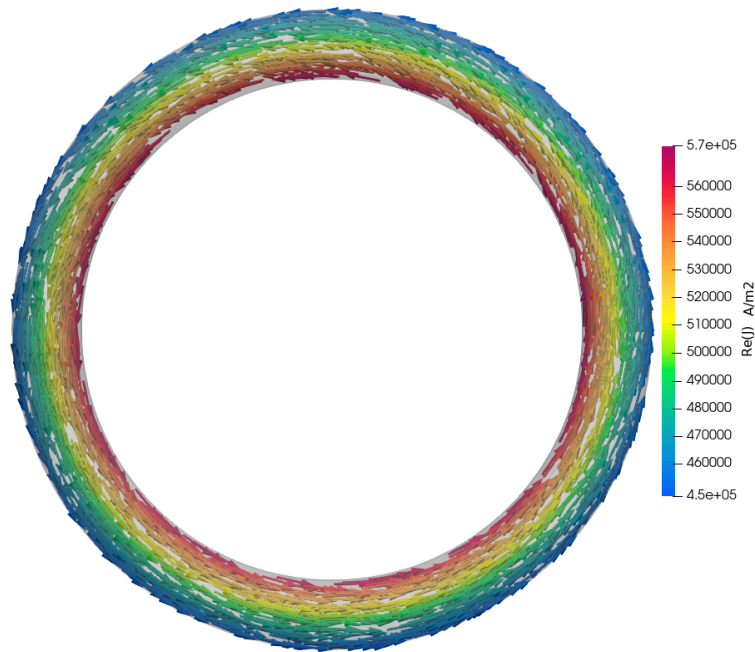


The same, obliquely, so the tube reads as a tube. Flat-shaded: 3D lighting darkens one side by more than Φ changes across a colour band.

This is the picture of what an internal cut actually does, and nothing in cases 01–04 looks like it. One side of the cut is the port's 0 V reference; the other floats to whatever the solve returns. **That jump is the terminal voltage** — $3.543192\text{e-}04 \text{ V}$ here, for 1 A, which is the impedance.

Φ is single-valued on each side: 193 nodes sit exactly on the cut plane and every one reads the same value, so the jump is a genuine discontinuity across a zero-thickness interface, not a gradient through a thin region.

3. The current



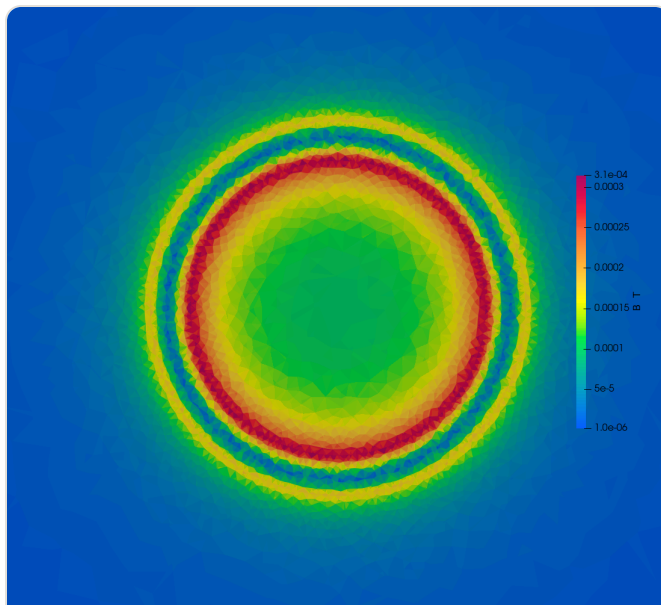
$\mathbf{J} = \sigma \mathbf{E}$ on the ring, every 11th cell, coloured by magnitude. The arrows close the loop.

Look at the colour, not just the arrows. $|\mathbf{J}|$ is highest on the **inner** radius — $5.8e5$ against $4.4e5$ A/m², a ratio of **1.32**. That is not noise: current flowing azimuthally takes a shorter path on the inside, so it crowds there. $(R+a)/(R-a) = 1.28$, and the measured 1.32 is that effect.

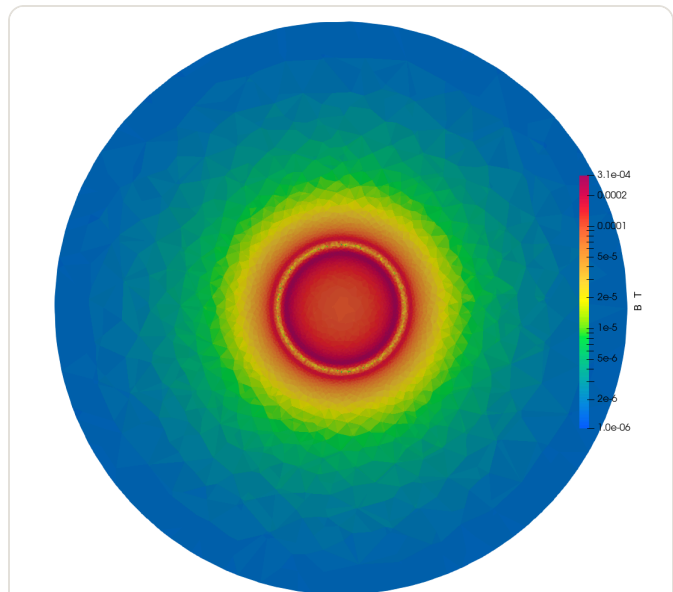
It is also why the exact R below is **not** $2\pi R/(\sigma A)$.

4. The fields

4.1 B in the xy plane



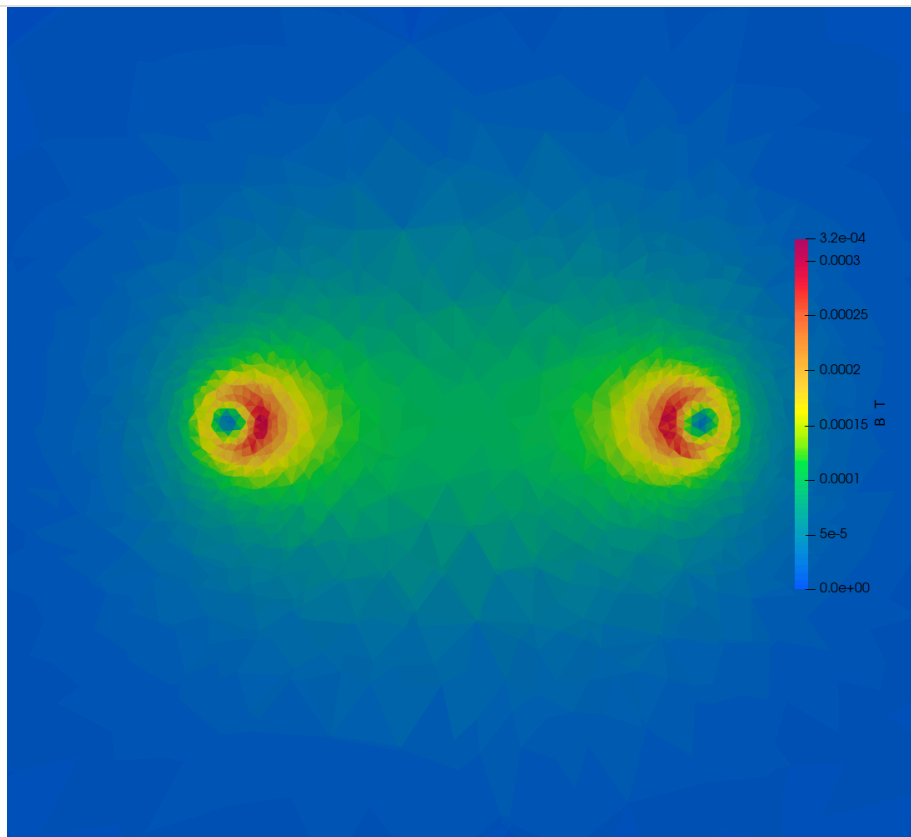
|B| through the tube centreline, zoomed. Red at the **inner** tube surface, a minimum at the tube axis, yellow at the outer surface.



The whole 30 mm domain, log scale, so the decay to the wall is visible.

Three things in the zoomed view are worth naming, because each is a check. **|B|** has a **minimum at the tube axis** — the wire's own field vanishes there. It **peaks at the tube surface**, where Ampère gives $\mu_0 I / (2\pi a) = 2.5e-4 \text{ T}$. And the inner surface is **hotter than the outer**, because the rest of the loop's field adds on the inside and subtracts on the outside.

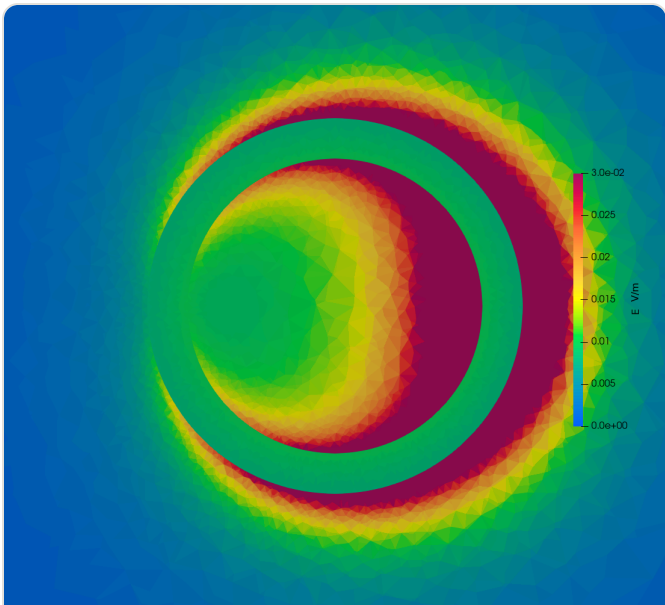
4.2 The port plane



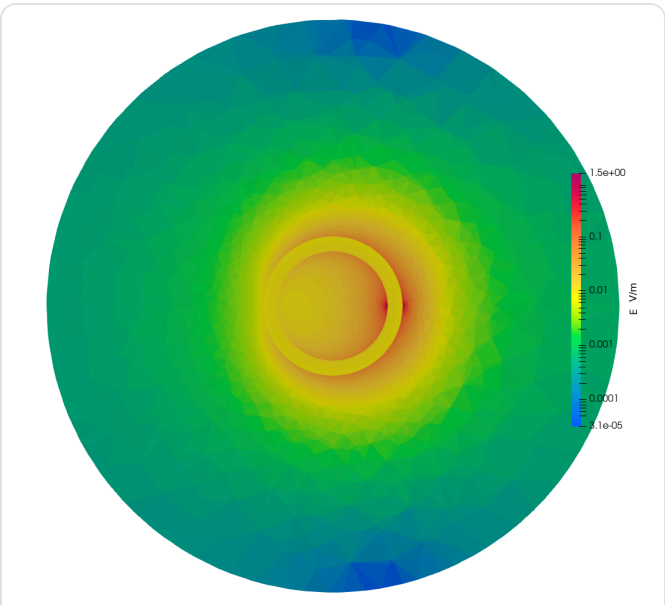
$|B|$ on the $y = 0$ plane, which cuts the tube twice — at the port ($\theta = 0$, right) and at $\theta = \pi$ (left). Range pinned: see the note in §4.3.

Each conductor carries its own circulating field with a minimum on its axis, and the loop's hole between them is brighter than the outside — the two contributions add in there and oppose outside. This is the plane the cut is made on.

4.3 E in the xy plane, and a warning about colour ranges



|E|, zoomed, range pinned to 3.0e-02. The conductor is the uniform green annulus; the air around it is *higher*.



The whole domain, log scale.

MEDIAN E 	
inside the conductor	8.584e-03 V/m ($J/\sigma = 8.621e-03$, 0.4 %)
air within 3 mm of the tube	2.037e-02 V/m
air beyond $\rho = 25$ mm	3.719e-04 V/m
at the cut itself	1.72 V/m — 200x the conductor

The air's E is larger than the conductor's — and so it is in every other case. An earlier version of this report called that surprising and implied it was something the ring did. It is not. Measured across the suite:

CASE	CONDUCTOR E 	AIR WITHIN 2.5A	RATIO	GLOBAL MAX / CONDUCTOR
02 rod, 1 V	24.9 V/m	55.7	2.2x	7.7x
03 rod, 1 A	2.45e-03	5.47e-03	2.2x	7.7x
04 rod, 393 Hz	1.51	9.84	6.5x	17.7x
05 torus, cut	8.58e-03	2.10e-02	2.4x	200x

What IS different is the dynamic range, and it is the cut that causes it. The ratio of air to conductor is ordinary here — 2.4 against the rods' 2.2. But the ratio of the frame's *maximum* to the conductor is **200x**, against 7.7x for a rod, because Φ jumps across a zero-thickness interface and the adjacent air sees 1.72 V/m.

That is also why the conductor reads BLUE here and deep red in case 02. It is the colour range, not the physics. Case 02's `06_E_in_wire.png` thresholds to the *wire only* and auto-ranges to the wire's own values, so the wire spans the whole colour map and tops out red — that figure's own report warns it magnifies noise in the fifth digit into a full rainbow. The figures here keep the air in frame and must span the cut spike, so the conductor lands at the bottom of the scale. Same field, different window.

The air's field is **electrostatic**, not `-j $\omega$ A` — that term is only $2.6\text{e-}04$ V/m here. The loop carries 0.354 mV from one side of the cut to the other and that voltage appears across the surrounding air, strongest near the cut where the gradient is.

Two ranges are pinned, and both had to be. `|E|` spikes to 1.72 V/m *at the cut*, because Φ jumps across a zero-thickness interface — on an auto range that one spike takes the whole colour map and everything else reads as flat blue, which is how this figure first came out. The port plane in §4.2 has the opposite problem: the slice spans the whole 30 mm domain while the view is zoomed to 12 mm, so an auto range is set by a far-field minimum three decades below anything visible and the picture lands uniformly red.

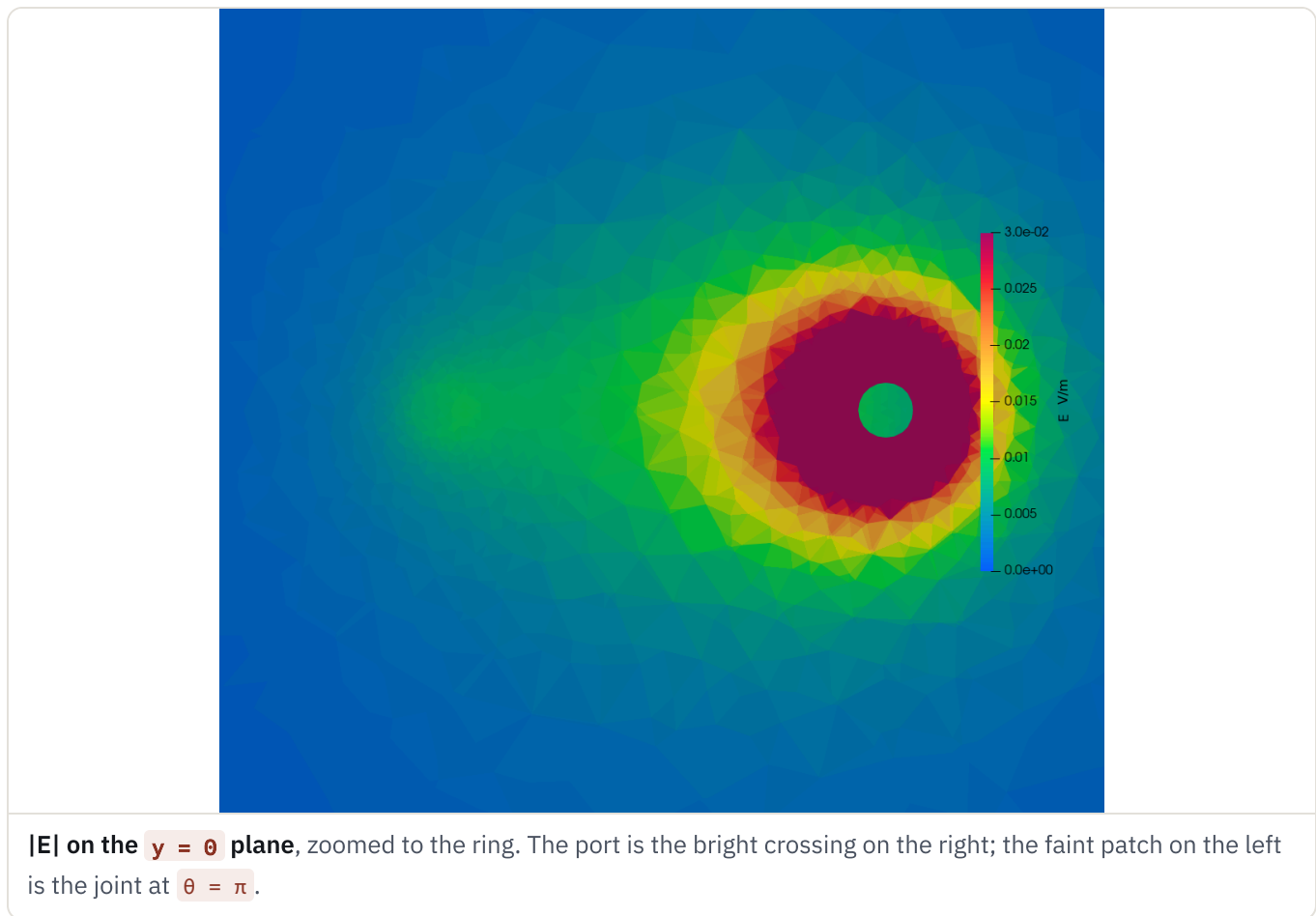
4.4 E on the plane normal to the torus, through the port

“Normal to the torus” at `$\theta = 0$` means the plane whose normal is the centreline tangent there, which is `+y` — so it is the `y = 0` plane, and **that is the cut plane itself**. It slices the tube twice: at the port on the `+x` side, and at `$\theta = \pi$` on the `-x` side.

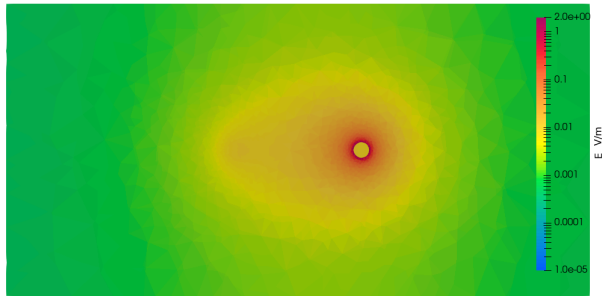
How to read these. You are looking along `-y` at a *vertical* plane containing the `x` and `z` axes. **+x is to the right.** The plane crosses the tube twice — at `x = +6.5 mm`, which is the **port**, and at `x = -6.5 mm`, which is **just a joint** where the two half-rings meet with no potential difference across them. Only one of those two is a source, which is why only one is bright.

CROSSING	CONDUCTOR $ E $	AIR RIGHT AROUND IT
$x > 0$, $\theta = 0$ — the port	0.008713 V/m	median 0.174 , max 1.72 V/m
$x < 0$, $\theta = \pi$ — a plain joint	0.008703 V/m	median 0.0076, max 0.0099 V/m

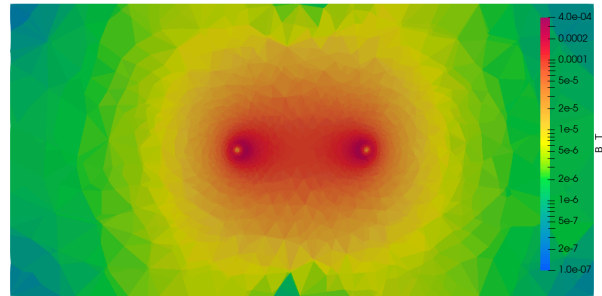
Inside the two conductors $|E|$ is the **same** — 0.008713 against 0.008703 V/m, both J/σ , because the same current flows through both. Outside them it differs by **23x**. That difference *is* the port.



The same cut over the whole domain



|E|, the full 60 × 30 mm cut. Log scale over five decades: 1.7 V/m at the cut face down to below 1e-4 at the wall.

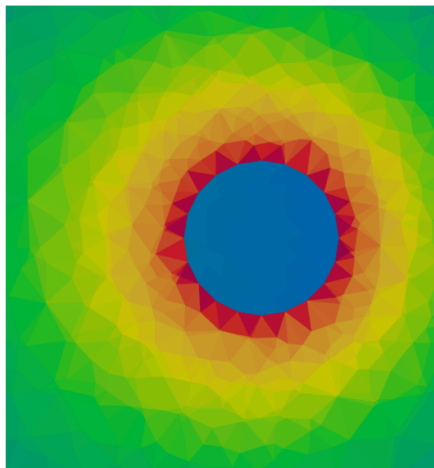


|B| on the same cut, for comparison. It is **continuous** across the port and shows both crossings equally — the asymmetry above is an **E** effect only.

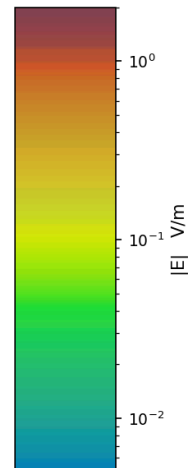
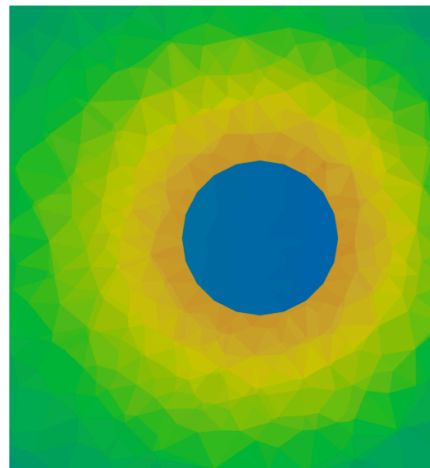
The **B** figure is the control. A magnetic field does not care that Φ jumps, so both tube crossings look alike there, and the two pictures side by side separate what the port does to **E** from what the geometry does to everything.

|E| at the port cross-section, same view and same log scale

ON the cut ($\theta = 0$)
air around the conductor reaches 1–2 V/m



one tube radius OFF the cut
the same air peaks near 0.2 V/m



Blue disc = the conductor, uniform at $J/\sigma = 8.6\text{e-}03$ V/m (no skin effect at $a/\delta = 0.086$). The difference between the two panels IS the port: Φ jumps 0.354 mV across a zero-thickness interface on the left, and is continuous on the right.

The measurement is the difference between these two. Same view, same tube, same log scale — the left panel lies *on* the cut, the right one tube radius off it. The red ring on the left is the port.

Each panel alone is an unremarkable blob, which is why they are drawn in one frame rather than merely adjacent: the eye has to compare them, and a shared colour bar is the only way that comparison means anything. The blue disc is the conductor, uniform at J/σ

because $a/\delta = 0.086$ leaves no skin effect to resolve. `make_port_pair.py` composes it, sampling ParaView's own colour ramp out of the render rather than guessing a matplotlib equivalent.

This slice lies exactly on a discontinuity, which is the point. Φ jumps across the cut, so $E = -j\omega A - \nabla\Phi$ does too: the whole 0.354 mV appears across a zero-thickness interface, and the adjacent air sees an enormous gradient. B is continuous there and does not care, which is why §4.2's plot of the same plane needs none of this care.

AIR SHELL HUGGING THE TUBE, $+X$ SIDE	MEDIAN $ E $
at the cut, $ y < 0.3a$	0.4207 V/m
one tube radius off, $0.7a < y < 1.3a$	0.1708 V/m
two to three radii round	0.0999 V/m
inside the conductor	0.008584 V/m (J/σ)
global maximum, at the cut face	1.7218 V/m — 200x the conductor

The field falls by 2.5x within one tube radius of the cut and by 4.2x by two or three. Both zooms are **log scaled**: the view spans more than two decades, from the conductor's 8.6e-03 to 1.7 V/m, and a linear range that resolves the conductor saturates every air cell around it into one flat colour.

5. Results

5.1 R — 0.11 %

measured	3.543192e-04 Ω
exact for this shape	3.546955e-04 Ω
error	−0.106 %

The exact value is not $2\pi R/(\sigma A)$. Current flows azimuthally, so the path length is $2\pi\rho$ and varies across the section: the conductance is $\sigma \int dA/(2\pi\rho)$, and the current crowds to the inner radius as §3 shows. Ignoring that is a **0.44 %** error — four times the solver's own.

`verify_loop.py` takes it straight off the mesh as $(1/2\pi) \int dV/\rho^2$, assuming nothing about the shape.

The 2.5 % gap to the reference tool decomposes exactly: **+1.61 %** because our cross-section is a 20-gon, 1.64 % less area than a circle, and 0.91 % between the reference and the exact round-torus value.

5.2 L — still 2 % out, and it is ours

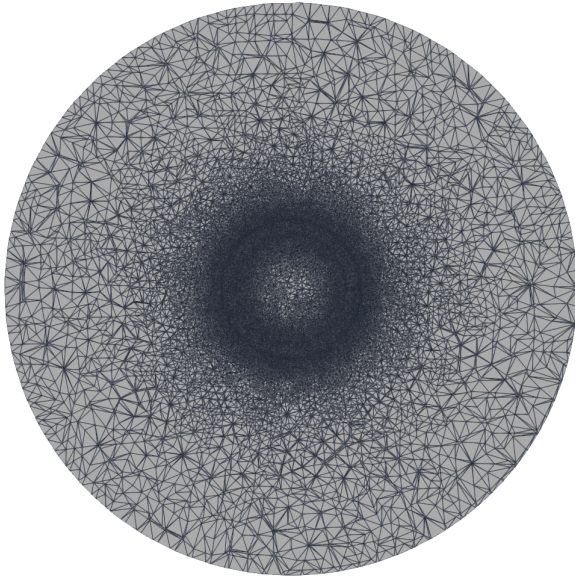
	L	VS THE CLOSED FORM
measured	19.377 nH	-2.15 %
reference tool	19.748 nH	-0.27 %
closed form, free space	19.8027 nH	—

The reference agrees with theory; we do not. **It was -4.02 %, and half of that turned out to be a mesh bug** — see §6.2. The remaining 2.15 % is open.

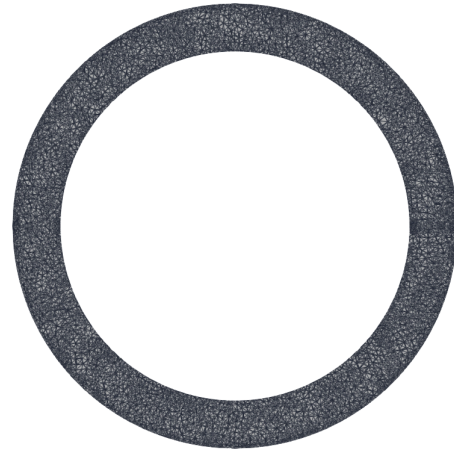
The domain is ruled out: doubling it to 60 mm moved `L` by +0.2 %, and a dipole estimate puts the field energy beyond 30 mm at about 0.09 nH, 0.4 %. The remaining suspect is resolution of the **air beyond one wire radius**: a wire loop's magnetic energy is logarithmically distributed, roughly half of it between 1 and 10 wire radii from the surface, and that shell is meshed at 0.25 mm growing to 4 mm. Refining it needs about 700k tets in the shell alone, which does not fit in 32 GB. **Recorded rather than resolved**, and `expected.txt` pins `L` as a regression guard rather than a validation.

Extracting L here is ill-conditioned. $\omega L/R = 0.017$: the reactance is 1.7 % of the resistance, so `Im(Z)` is a small difference on a large product. Dividing by the *measured* current instead of the imposed 1 A lets that current's own 0.02 % imaginary part into the answer and moves `L` by 1.3 % — fifty times its own size. The verifier divides by the imposed current.

6. The mesh, and two bugs found building it



Whole domain, graded from 0.25 mm at the ring to 4 mm at the wall.



The ring, about 5 elements across the 1.6 mm tube.

6.1 An OCC Torus will not mesh with a cut in it

The obvious construction — an **OpenCASCADE Torus**, cut, fragmented against the air — fails with *“Invalid boundary mesh (overlapping facets)”*: a curved torus face overlapping itself. That was tried **eight ways**: partial tori welded with **Coherence**, cuts by oversized discs, by exact-size discs, by thin boxes, cuts moved off the parametric seam, the seam rotated away from the cuts, the air fragmented before and after the cut. Every one failed identically.

Revolving a polygon has none of that trouble, and it is what this project does for every other conductor anyway, because planar lateral faces make $\partial\Phi/\partial n = 0$ hold exactly.

Note also that a ring needs **two** cuts. Cut in one place it is still one connected volume, because the current can go round the other way.

6.2 One $+\pi$ sweep and one $-\pi$ sweep is wrong in a way that is easy to miss

The ring was first revolved as one **+Pi** extrude and one **-Pi** extrude. It meshes, it passes every check, and **both halves come out with the right volume** — 40.386 against 40.307 mm³, equal to 0.2 %. But:

	TETS	MEAN H	VOLUME
+Pi half	11,621	0.302 mm	40.386 mm ³
-Pi half	4,557	0.373 mm	40.307 mm ³

Half the conductor was quietly meshed two and a half times coarser. It showed up as a ragged lower half in the field plots, and was measured before it was believed. Chaining two *positive* sweeps — the second starting from the first's end face — gives a count ratio of 0.988.

And it moved the physics. **L** went 19.038 → 19.377 nH, closing 1.87 of the 4.02 points of the gap to theory. It also **retracts** an earlier conclusion recorded in this case: that refining the tube moved **L** the wrong way, so resolution was not the cause. That test refined **lc_ring** globally while half the ring stayed coarse, so it was measuring the wrong thing. Resolution *is* part of it.

7. What is asserted automatically

check.bat runs this case through **verify_loop.py**, not **../verify.py**: the rod verifier finds the drive by differencing Φ between two end caps, and a ring has none. A case-local **verify_*.py** wins where one exists.

 I through the cut	0.9999784 vs 1.0, tol 1e-3
 I at $\theta = \pi/2$	1.000183 vs 1.0, tol 1e-3
R vs exact for this shape	0.99894 vs 1.0, tol 2e-3
L	19.377 nH, tol 0.05 — a regression guard , not a validation

Reproducing

```
gmsb loop.geo -3 -o loop.msh
..\run_case.bat loop_50hz.aphi
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" make_plots.py output
"C:\Program Files\ParaView 6.1.1\bin\pvbatch.exe" verify_loop.py
sh build_doc.sh
```

`loop_big.geo` is the 60 mm domain used for the boundary test in §5.2; it is kept because a negative result is worth being able to reproduce. Run `build_doc.sh` after regenerating **figures**, or this document keeps showing the previous run's pictures — they are baked in.